

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We investigate how physical extent and population magnitude condition the emergence of complex behavior in open-ended ecological environments. In our setting, agents remain unsupervised and receive neither explicit rewards nor externally specified learning objectives; instead, they undergo evolutionary change through reproduction, mutation, and selection. Through their actions, agents reciprocally transform both the environment and the surrounding population, thereby participating in a continually reconfigured dynamic ecology. Our objective is therefore not the optimization of a single high-performing policy, but the characterization of how behavior arises, persists, and changes across large populations under natural competition and environmental constraint. Using modern accelerators together with a new multi-agent simulator, we scale environmental area and population size beyond previously attempted regimes, reaching more than 60,000 agents, each endowed with its own evolved neural-network policy. Across these settings, we identify a range of emergent behaviors, including long-range resource extraction, vision-mediated foraging, and predation, that arise from competitive and survival pressures. We further examine how sensing modalities and environmental scale modulate the appearance of these behaviors, finding that some behaviors arise only in sufficiently large environments and populations, and that larger scales increase the stability and reproducibility of emergent behavioral patterns. Although evolutionary computation has a long and substantial history, our scaling results on contemporary hardware suggest promising directions for treating ecology as an instrument of machine learning in an era characterized by increasingly abundant computation and efficient machine-learning frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been examined across a broad range of biological disciplines, including evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Accounts of these emergent dynamics extend from theories of speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) to formal analyses in population genetics (Fisher 1930; Wright 1931).

Across these fields, decades of mathematical modeling, laboratory experimentation, and field observation have substantially refined our understanding of the mechanisms governing natural evolution. Nevertheless, the direct study of emergent behavior in natural ecosystems remains constrained by formidable practical and ethical obstacles: as ecosystems increase in size and complexity, they become progressively more difficult to control, instrument, and measure, while controlled interventions in large-scale natural settings may damage or displace wild populations.

To investigate the emergence of complex behavior under competitive and environmental pressures, we study the open-ended evolution of neural-network policies in large-scale ecological simulations. Within this setting, agents are not assigned an explicit objective or reward signal; rather, they acquire resources in order to survive and reproduce according to the dynamics of the environment. Policy variation enters only through mutation, as parents transmit modified versions of their policies to their offspring. Given the rapid expansion of computational resources available to researchers, our aim is to examine open-ended evolutionary dynamics in populations of simple digital organisms at scales

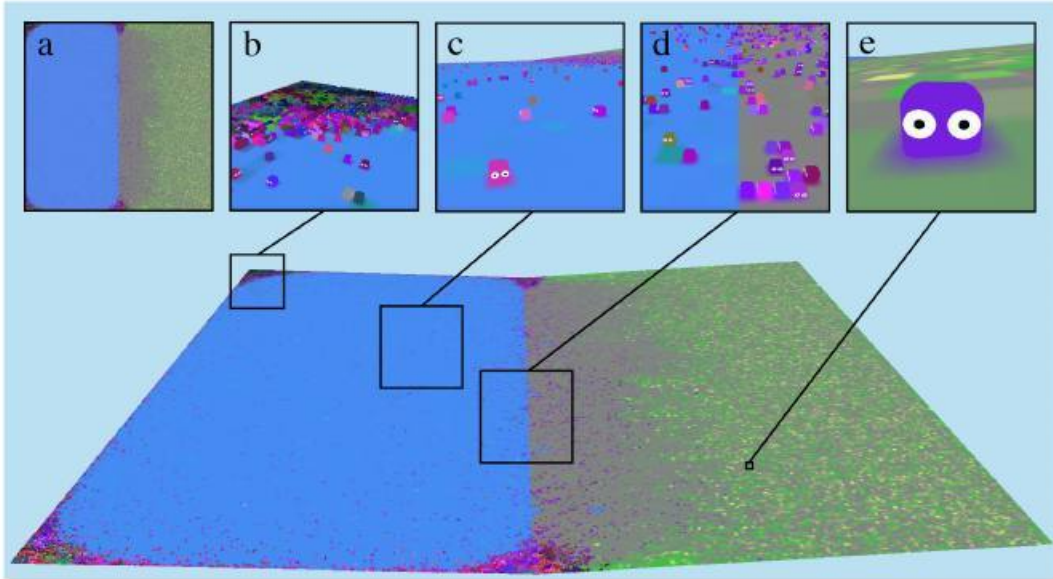


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) presents a top-down view of the entire map, (b) shows agents clustered in the corner, (c) shows agents in the water, (d) shows agents on the sand, and (e) shows agents mining for resources at a distant corner.

that were previously impractical, thereby providing new tools for analyzing how environmental extent and population size shape behavioral emergence.

To conduct this study, we use a new JAX-based simulation environment that enables rapid evaluation of large grid worlds. Our largest experiments contain more than 60,000 individual agents distributed over physical areas exceeding 1,000,000 grid cells. Agents in this environment must navigate in order to locate food and other resources required for survival, and they may reproduce after accumulating sufficient resources to generate a mutated copy of themselves. Figure 1 provides a 3D visualization of this environment.

Our goal is not only to clarify ecological phenomena through simulation, but also to examine ecology as a mechanism for producing machine intelligence. Recent progress in AI has demonstrated that scale does not merely improve models; it can qualitatively transform their capabilities (Wei et al., 2022a). Despite this observation, many attempts at embodied learning have remained confined to comparatively small environments with limited population sizes. Our work therefore asks what capabilities emerge when scale is applied not to a single model alone, but to an entire ecosystem. Just as reasoning abilities may appear in individual models only beyond particular size thresholds (Wei et al., 2022b), we seek to understand which strategies evolve in populations only after environmental size and complexity become sufficiently rich.

Through extensive experiments, we identify multiple cases in which sensor configurations and large environmental scales substantially influence the behaviors produced by simulation, with consequences that persist over long ecological horizons. For example, we show that agents equipped with simple compass sensors can adapt to perform long-range resource-gathering expeditions, although this behavior emerges less reliably at small scales. We also show that agents with vision sensors evolve more effective foraging and attacking behavior than agents without vision, and that these effects again appear more consistently in larger environments. Collectively, these experiments indicate that the scale and complexity enabled by modern hardware accelerators create promising opportunities for studying how ecological evolution can generate rich behavioral repertoires.

2 RELATED WORK

The emergence of life has motivated computational research since the field’s inception. Conway’s Game of Life (Games, 1970) demonstrates how population-level phenomena can arise from simple

local update rules, while early artificial-life systems such as Tierra (Ray 1992) and Avida (Brown 1987) showed that local motion rules can generate emergent flocking and schooling behaviors.

Researchers have further used these principles to evolve the locomotor morphology (Sims, 1994a,b; Ha 2018; Gupta et al. 2021; Auerbach et al. 2014; Auerbach & Bongard 2012; Jeon et al. 2025; Auerbach & Bongard 2014; Szerlip & Stanley 2013) and sensing capabilities (Pratt et al. 2022; Mugan & MacIver 2020; Tiwary et al. 2025) of virtual agents, while neuroevolutionary methods have directly evolved neural-network architectures (Stanley & Miikkulainen 2002, 2004; Stanley et al. 2009). In contrast with these approaches, we do not seek to produce agents that adapt more effectively to their environments through a reward or objective signal; rather, we examine which behaviors arise spontaneously under ecological pressure and how environmental scale and complexity modulate those behaviors.

Evolution has also informed the internal dynamics of optimization algorithms. Genetic algorithms (Holland 1992; Jong 1975) and evolutionary strategies (Wierstra et al. 2014; Hansen & Ostermeier 2001; Rechenberg 1973; Schwefel 1995; Conti et al. 2018) translate principles of evolution and natural selection into optimization procedures, while related methods use evolutionary mechanisms to encourage novelty among discovered solutions (Lehman & Stanley 2011; Cully et al. 2015). Other work has applied these approaches to optimize neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al. 2017; Such et al. 2017; Mania et al. 2018). Although our agents are updated through evolutionary mechanisms, our focus is the analysis of emergent behavior rather than the development of a new optimization algorithm or the application of an existing optimizer to a fixed objective.

A related tradition draws biological inspiration to develop organisms within simulated environments. Early examples include Bedau (1992), Yeager et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). More recent work uses cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al. 2020; Robinson 2021; Heinemann 2021; Randazzo & Mordvintsev 2023; Caussan 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to investigate the emergence of language and communication among agents. Emergence through autocurricula (Leibo et al. 2019), in which populations of agents create increasingly challenging learning landscapes for one another, has likewise been examined in environments with smaller populations than ours (Leibo et al. 2018; Team et al. 2021). Additionally, Wang et al. (2019) and Aki et al. (2024) invert the usual formulation by evolving environments specifically optimized for agent learning. Although these works often evolve the environment rather than the agent, they motivate our study by showing how environmental complexity can elicit sophisticated behavior.

Most closely related to our work are efforts that simulate interacting agent populations within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to use environmental tools to evade seeker agents, with both populations trained through reinforcement learning. Other systems similarly train populations for specific tasks, including video-game play (Jaderberg et al. 2019), object manipulation (Petrenko et al. 2021), and agent capture (Zheng et al. 2017; Yamada et al. 2020). Although many of these works investigate emergence, they do so within fixed tasks and objectives. The Neural MMO framework (Suarez et al. 2019, 2023) adopts a more open-ended reinforcement-learning formulation by placing agents in a massively multi-agent game and optimizing for survival rather than for narrowly specified objectives. However, because the population is capped at a fixed number of agents, with experiments examining up to 128 concurrent agents and without generational inheritance, its ability to probe population dynamics remains limited. Hamon et al. (2023) is closer to our setting through variable population sizes, genetically inspired evolution, and generational timelines, although its environment is substantially smaller, less feature-rich, and does not directly examine the importance of scale. Similarly, Lu et al. (2024) simulates agents evolving to gather resources. This work, however, also operates at much smaller scale, with a maximum of 256 agents, and emphasizes tool use through programmable robots. While these prior systems take important steps toward the study of emergent ecological dynamics, a comprehensive account requires populations that can grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can emerge. Our work integrates these elements, enabling examination of ecological phenomena inaccessible in prior work.

Finally, we are also motivated by recent work emphasizing environment speed and scale through optimized libraries that increase computational throughput and the number of agents able to interact in game-like environments (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). We build on this progress in computational efficiency, using related JAX-based acceleration techniques to achieve large-scale, high-throughput simulation in ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interactions between agent populations and their environment as a multiplayer game class that we call an **Ecological Game (EG)**. Analogously to Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al., 1998 and Albrecht et al., 2024 for reference), an EG retains underlying Markovian transition dynamics while being structured to represent changing populations in an objective-free setting. As in an SG, an EG includes a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and the set of player actions $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Unlike an SG, however, it removes the reward function and introduces a population function $\Gamma(s_t)$, which specifies which agents in I are currently alive and records parental identity so that birth and death events can be inferred. Just as MDPs and SGs admit partially observable analogues, POMDPs and POSGs, an EG may also be partially observable, in which case we refer to it as a POEG. A POEG differs from an EG by withholding the full state from each agent and instead using an observation function $O(s_t, i)$ that maps a state-agent pair to that agent’s observation. Our environment is a POEG whose observation function varies with the sensor configurations described in the next subsection.

3.2 ENVIRONMENT

The POEG instantiated in our experiments is a large grid world populated by agents that must acquire multiple resources in order to survive and reproduce. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, where s_t^m denotes 2D map-associated data, including terrain and resource distributions, and s_t^a denotes agent-associated data, including position, health, and internal resource stores.

Terrain: Each map contains a rock layer that defines the height of every grid cell. Water is initially assigned to cells whose rock value falls below a preset sea level, and during an episode it subsequently flows downhill according to a discrete flow simulation determined by the rock-induced height field. In the experiments below, we use the six global terrain configurations shown in Figure 2: **Ocean, Beach, Island, Lake, Isthmus, and Channel**. Experiments are conducted at 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grid sizes; when the grid is enlarged or reduced, the initial population is scaled in proportion to total map area, after which population size fluctuates endogenously according to environmental dynamics.

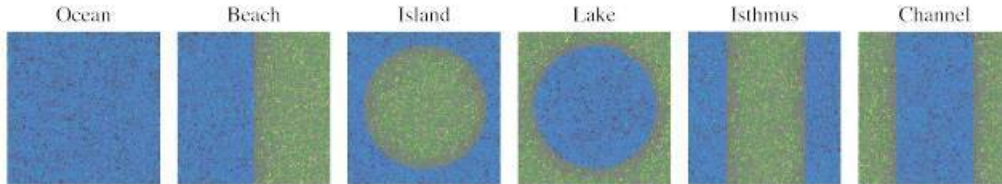


Figure 2: Overhead images of the terrains used in our experiments at the 256×256 grid size. Small dark spots denote individual agents, whereas green and yellow indicate concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most aquatic biomass had been consumed but before agents had evolved the capacity to persist on land for extended periods.

Resources and Agent Health: Agents must acquire **water**, **energy**, and **biomass** in order to persist and reproduce in these environments. At initialization, biomass and energy are randomly distributed across the world, but both may subsequently be consumed, transported, and redistributed by individual agents.

Every agent action consumes a small quantity of energy and water. Expended water is returned to the landscape, where it flows downhill toward the nearest body of water, whereas expended energy is depleted and destroyed. Biomass serves as the material basis for reproduction: because every agent requires a fixed amount of biomass simply to exist, reproduction is possible only after sufficient excess biomass has been accumulated for transfer to offspring. As with water, total biomass is conserved throughout the simulation; unlike water, however, biomass neither flows downhill nor moves autonomously. Energy is not conserved, but instead grows slowly on free-standing biomass. These resource dynamics were designed to make the ecological problem modestly richer than single-food-source collection. For ablations on resource rules, see Appendix F.2

Agents additionally possess health points (HP), which must be maintained to avoid death. Over-spending a critical resource causes rapid health deterioration, although agents can gradually restore HP over time by using energy and water. Agents may also attack one another, thereby reducing the target’s HP, and they incur a small but increasing age-dependent damage term, so that old age ultimately causes death in the absence of other lethal events. When an agent’s HP reaches zero, any resources it was carrying are deposited back into the environment at its current grid cell. Consequently, an agent that attacks and kills another agent may subsequently consume the victim’s resources, producing a basic form of predation. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents observe the environment through four sensor types. The **internal** sensor reports age, health, and carried resources; the **external** sensor detects resources occupying the same grid cell as the agent; a **compass**, inspired by magnetoreception in birds and other animals, encodes the agent’s global orientation as a 4-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, with three color channels and a fourth channel representing local elevation differences. For ablations on vision configurations, see Appendix F.6

Actions: Agents choose from a finite discrete action set organized into five categories. The first category, **Rest**, produces no movement or environmental change but carries the lowest energy cost. Three **Move** actions allow an agent to rotate clockwise, rotate counterclockwise, or advance one grid cell; because cells cannot contain multiple agents, attempted movement into an occupied cell, or simultaneous movement by multiple agents into the same cell, causes the relevant movement action to be canceled. Agents also have a single **Attack** action, which kills all agents in the 3×3 square directly in front of the attacker. For ablations on attack strength, see Appendix F.7. Finally, three distinct **Eat** actions correspond to the three resource types. When an agent occupies a cell containing a given resource, the corresponding eat action transfers a fixed quantity of that resource from the environment into the agent’s stomach, which has a fixed maximum capacity.

Finally, agents possess a single **Reproduce** action that creates offspring through single-parent reproduction. This action succeeds only when the parent has accumulated sufficient resources for transfer to the child. The newly instantiated agent is placed directly behind its parent, receives a mutated copy of the parent’s policy weights, and immediately inherits a fixed quantity of the parent’s resources.

In the experiments below, we frequently disable particular sensors or the attack action in order to isolate the ecological consequences of different agent capacities. We consider three sensing configurations: internal+external resource sensing only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). We append **+A** when attacking is enabled; for instance, (**RC+A**) denotes an agent with resource sensors, a compass, and the ability to attack.

3.3 POLICIES

Agent policies are implemented as small, memoryless MLPs. We use memoryless agents to simplify the computational model and reduce the number of network parameters. Raw values from each sensor are flattened, normalized to the range $[-1, 1]$, and passed through separate linear layers with output dimension 64. The resulting sensor embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of available actions. Actions are sampled from a softmax distribution whose temperature is itself allowed to evolve with the network weights. To support larger populations and faster execution, weights are represented using half-

precision bfloat16 values. Depending on the active sensor set, each agent network contains roughly 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5

Across all runs, weights in the initial agent population are assigned using Kaiming initialization (He et al. 2015), while bias vectors are initialized to zero. Policies for agents born during an episode are produced by copying the parent’s weights and adding Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation procedure resembles mechanisms used in prior neuroevolution work (Such et al. 2017). Although more elaborate mechanisms, including crossover, are possible, we use this simpler reproduction scheme in order to isolate the effects of environmental scale in a minimally complicated setting. Empirically, the use of low-precision floating-point values required a larger mutation rate. We report a mutation-rate ablation in Appendix F.4

Agents also possess a 3-channel trait that controls their rendered color on the map. This color trait mutates in the same manner as the network weights and determines an agent’s appearance when observed by other agents through the vision sensor. In principle, this mechanism permits agents to discriminate among different visually distinguishable agent types.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al. 2018) and executed on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise specified, each experiment is run for 2M environment steps or until extinction, with per-seed wall-clock times ranging from less than one hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E provides additional detail on simulator architecture, reports further scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

The following experiments examine how sensing modality, environmental scale, and population size jointly shape the emergent behavior of agents.

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment investigates whether agents can acquire the capacity to travel long distances inland in search of resources. We show that, in environments with clear directional structure, agents equipped with a compass (**RC**) consistently develop substantially more effective long-range resource-gathering behavior than the baseline (**R**) agents, which lack a compass. We further find that this behavior emerges more consistently and operates more effectively as environmental scale increases.

We begin with a visual example of the (**R**) agent class, shown in Figure 3. After 5000 time steps (a), all initially land-based agents have died because of water scarcity, while agents located in the water have rapidly proliferated due to abundant initial resources. At this early stage, the population does not yet contain policies capable of efficiently searching for resources. Following this initial population peak, aquatic agents occasionally wander onto land (b); when they do so, they often fail to return to the ocean and die, depositing their biomass on land. By 50,000 time steps (c), this gradual process has nearly doubled the quantity of uneaten terrestrial biomass. Because biomass has been transferred from water to land, the aquatic population contracts as insufficient biomass remains in the water to sustain a large population. By 125,000 steps (d), almost all biomass has been displaced onto land and the population is nearly extinct, with the final agent dying around 185,000 steps (e).

This example illustrates the ecological consequences of agents’ inability to navigate back to water. Individual deaths do not merely remove agents from the population; they also redistribute resources in ways that substantially affect subsequent population-level dynamics. Simultaneously, this ecological collapse creates an opportunity for any agent capable of discovering a policy that permits it to walk onto land, forage from the biomass deposits left by dead ancestors, and safely return to water.

In the next example, shown in Figure 4, agents are endowed with an internal compass (**RC**). Initially, a similar pattern appears (a): population size decreases while free terrestrial biomass increases

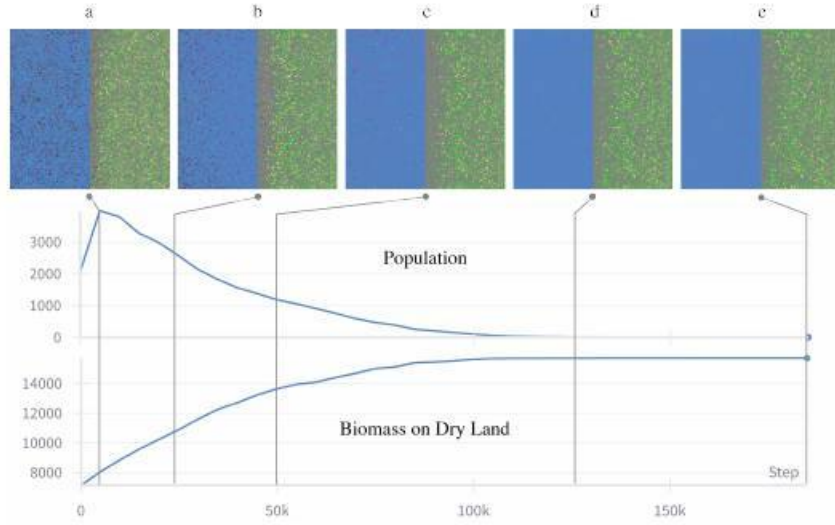


Figure 3: A representative trial with **(R)** agents in a 256×256 **Beach** environment. The upper chart reports population over time, whereas the lower chart reports free biomass on dry land.

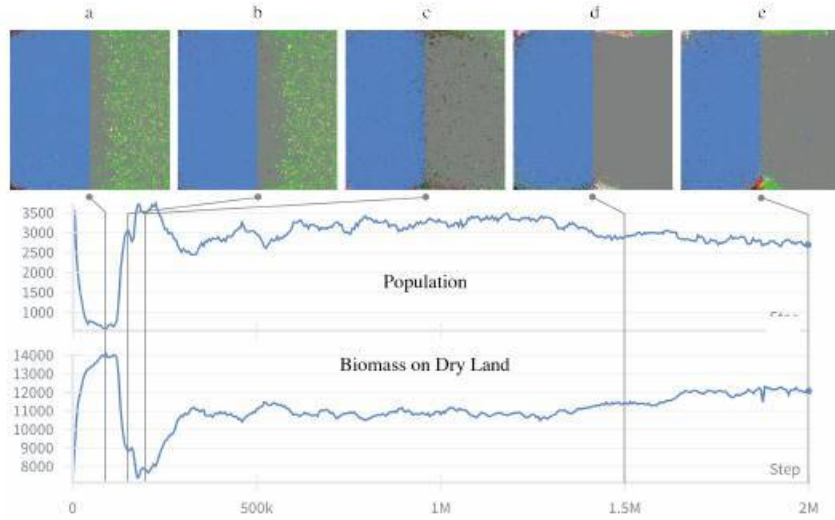


Figure 4: A representative trial with **(RC)** agents in a 256×256 **Beach** environment. The upper chart reports population over time, whereas the lower chart reports free biomass on dry land.

rapidly. In this run, however, some agents use the compass to drift into the relative safety of the corners. Around 150,000 steps (b), some agents begin to “mine” terrestrial biomass, using compass information to navigate repeatedly between land and water. By 200,000 steps (c), these miners have cleared the entire beach. Although the population subsequently undergoes minor spatial reorganization, with agents forming clumps along the edges (d), it remains largely stable through the full 2,000,000 time steps (e).

We quantify these observations by conducting a full set of **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each experiment is run with four random seeds, and we record whether mining behavior emerges, operationalized as a reduction of at least 10% in free dry-land biomass relative to its historical maximum. We additionally record the onset time of the mining event, defined by the local maximum in terrestrial biomass, together with the percentage biomass decrease. Table 1 summarizes mining and extinction events, while Table 3 in Appendix A

Table 1: Mining events in the **Beach** world for populations with a compass (**RC**) compared with populations lacking a compass (**R**), evaluated across five world sizes and initial populations.  denotes the number of seeds exhibiting a mining event, and  denotes the number of seeds in which the population went extinct before 2M steps. Table 3 in Appendix A provides seed-level data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

provides seed-level details. Relative to (**R**) agents, (**RC**) agents exhibit mining behavior much more consistently. Mining becomes increasingly reliable in larger environments and occurs in all seeds at and above the 256×256 scale, whereas smaller scales produce less stable populations and a greater likelihood of early extinction.

To examine these effects more precisely, we run additional seeds with (**R**) and (**RC**) agents at 512×512 world size (32 seeds) and 128×128 world size (64 seeds), using a Kaplan-Meier estimator to measure the probability of extinction and mining as functions of simulator time. We find that extinction is rarer and occurs later in larger worlds, whereas mining is more frequent and occurs sooner in smaller grids. We also fit a Cox proportional hazards model to compare the timing of mining discovery and extinction across world sizes and compass conditions, again finding that larger world size substantially reduces extinction rate and enables more rapid mining discovery than smaller worlds. Compass availability produces a similar effect. Full details are provided in Appendix B.

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for compass-equipped agents (**RC**) compared with agents without a compass (**R**) at the 512 × 512 grid size.  denotes the number of seeds exhibiting a mining event, and  denotes the number of seeds in which the population went extinct before 2M steps. Table 4 in Appendix A provides seed-level data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We conduct analogous experiments in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512 × 512 grid size with both (**R**) and (**RC**) agents, thereby testing how terrain configuration affects the emergence of mining behavior. Table 2 reports mining and extinction events for these experiments, and Table 4 in Appendix A provides detailed information for each run. As in the **Beach** setting, compass-equipped agents exhibit earlier and more consistent mining, as well as more stable populations with lower extinction rates. (**RC**) agents adapt more readily to the island and isthmus environments than to the lake and channel, where extinction occasionally occurs. We hypothesize that this difference arises because moving directly east or west reliably returns an agent to water in the island and isthmus terrains, whereas the channel and lake do not provide an equally simple navigational rule. The (**R**) agent class fails to adapt reliably in any of these environments, although it struggles least on the island. Figure 12 in Appendix D visualizes mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We next conduct experiments in the **Ocean** environment to determine whether agents can adaptively exploit vision sensors to locate free resources or prey on other agents. We compare (**RCV+A**)

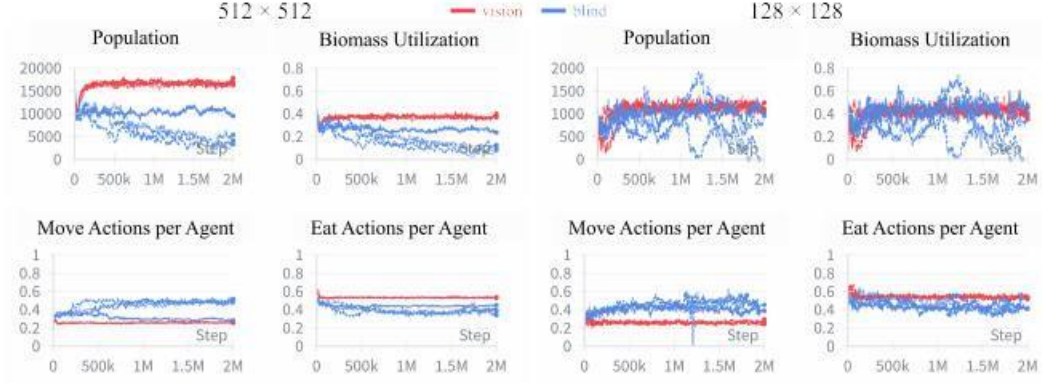


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with visual agents ($RCV + A$) (red) and four seeds with blind agents ($RC + A$) (blue). With vision, populations are larger and more stable, and biomass utilization is higher because agents move less while eating more frequently and effectively. At larger scales, these dynamics exhibit greater stability and lower across-run variance, suggesting that larger environments increase the advantage conferred by vision for adaptive foraging. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

agents, which have resource sensing, a compass, and a 7×7 visual sensor, against ($RC+A$) agents, which retain only resource and compass sensors.

Figure 5 reports population, biomass utilization, movement actions, and eat actions for agents in 512×512 and 128×128 grids. Additional grid sizes are shown in Figure 9 in Appendix C and summary statistics appear in Table 5 in Appendix A. Biomass utilization is computed as the percentage of total environmental biomass currently contained within agents. In episodes with vision sensors (red lines), agents spend less time moving and more time eating; these runs also support larger populations and achieve greater biomass utilization. Together, these dynamics indicate that agents adapt to use visual sensing for more efficient foraging. Across environmental scales, run-to-run variance decreases substantially in larger environments, and, unexpectedly, vision-enabled runs exhibit markedly more stable temporal dynamics, with reduced variation in these metrics over time.

Figure 6 reports averaged attacks per agent and homicides per attack for these runs. These statistics indicate that agents with vision attack relatively infrequently while maintaining a successful kill rate of approximately 75%. In large grids, blind agents sometimes also achieve high kill rates, although their dynamics are substantially less stable. Several confounding factors make the cause of this observation difficult to isolate. Through visual inspection (see Figure 13 in Appendix D), we find that high kill rates sometimes occur when agents accumulate in localized clusters, and that such clusters almost never appear in runs with visual agents. This pattern may indicate that visual agents can adapt to avoid one another. In smaller environments, the contrast between visual and blind agents is less pronounced, although visual-agent populations remain far more stable.

4.3 ABLATIONS

Appendix F presents ablations over key simulator hyperparameters, including initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8), in order to assess the sensitivity of emergent behaviors to simulator configuration. We additionally report experiments in Appendix F.3 that test whether the dynamics observed in larger environments can be explained merely by self-averaging.

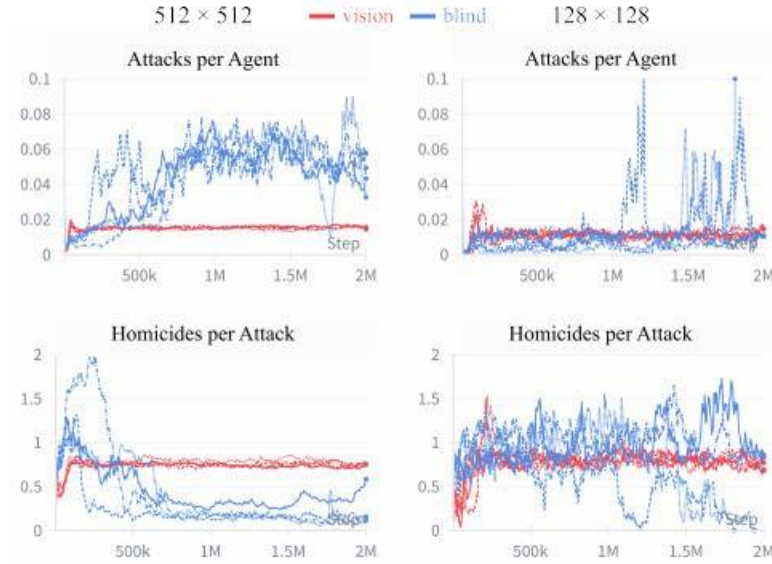


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with visual agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less often but with substantially higher precision, suggesting the emergence of specialized predation behavior. This dynamic is more pronounced at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

5 CONCLUSION

Our study underscores the promise of open-ended ecological settings as a generative paradigm for investigating the emergence of intelligence. Across experiments that vary both world scale and environmental complexity, we find that environmental extent and sensory richness jointly contribute to the development of more sophisticated sensing and decision-making policies. Taken together, these results suggest that ecological and evolutionary pressures, when coupled to richly structured environments, may provide a natural route toward increasingly capable artificial agents. Moreover, our experiments show that elaborate behaviors can be evolved in large populations within a practical hardware budget, without reliance on explicit objectives. We hope that these findings will motivate further investigation of ecology as both a mechanism for developing artificial intelligence and a framework for understanding how intelligent behavior can emerge.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world for populations with a compass (**RC**) compared with populations without a compass (**R**) across five scales.  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for (RC) agents compared with (R) agents. The grid size is 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision and Blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, measured as the percentage of total system biomass currently inside agents' stomachs. denotes move actions per agent, defined as the fraction of time an agent selected a move action, averaged across agents. Similarly, denotes the proportion of agent actions corresponding to eating. Reported values of , , , and are seed-level averages over all steps.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision and Blind agents in the **Ocean** world across scales.  denotes attacks per agent, defined as the fraction of time an agent selected the attack action, averaged across agents.  denotes homicides per attack, or the number of agents killed per attack action. Reported values of  and  are seed-level averages over all steps. The 0.562 value for  in seed 3 of the 128×128 (**RC+A**) run is an outlier; because this seed went extinct before 2M steps, dwindling population size produced a spike in attacks per agent.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare extinction rates and mining-emergence rates as functions of world size and agent access to global directional information through the compass sensor. Extinction is defined as the population reaching zero individuals, whereas mining behavior is defined as in the preceding experiments, namely a 10% reduction in free biomass on land.

Runs are grouped jointly by world size and compass access. We simulate 64 runs each for 128×128 worlds without a compass and 128×128 worlds with a compass, and 32 runs each for 512×512 worlds without a compass and 512×512 worlds with a compass. For each of the four groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that do not go extinct within 2M steps as right-censored. Figures 7 and 8 show that worlds tend to survive longer and produce mining sooner when they are larger or when agents have compass information.

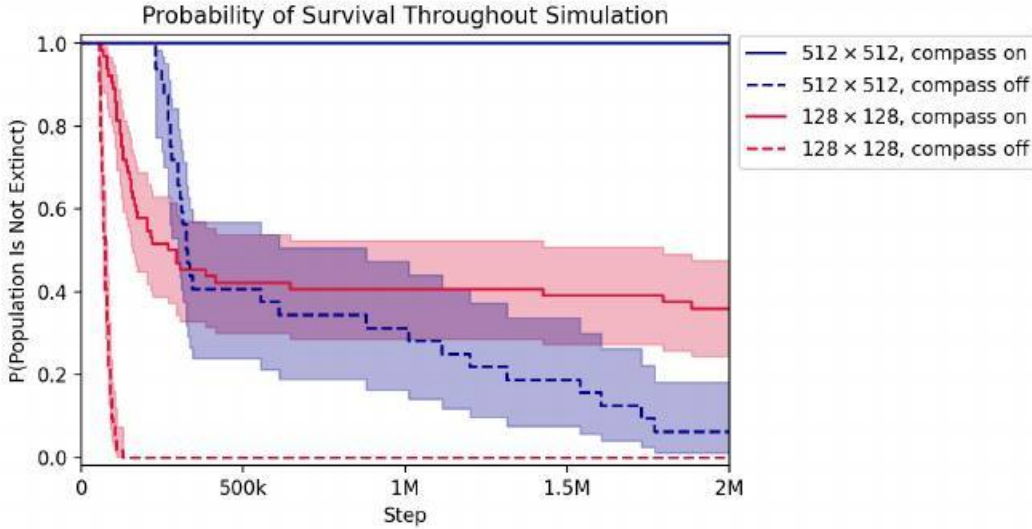


Figure 7: Probability that the population has not gone extinct, conditioned on world size, compass access, and timestep. We fit a separate non-parametric Kaplan-Meier estimator for each run group (e.g. 512×512 , compass on). Shaded regions denote 95% confidence intervals. Each 512×512 estimator is fit on 32 runs, and each 128×128 estimator is fit on 64 runs.

We compare emergence and extinction timing using a Cox proportional hazards model, a semi-parametric survival model that relates covariates to event hazard rates (Table 7). Runs that experience neither emergence nor extinction by the end of 2M time steps are treated as censored. For extinction, both larger world size and compass use substantially reduce hazard; the extinction rate in 128×128 worlds is approximately 25 times greater than the corresponding rate in 512×512 worlds ($HR=0.04$). Compass activation produces an effect of comparable magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior much more rapidly ($HR=6.07$), and compass use accelerates emergence dramatically ($HR=34.59$). The models exhibit good discriminative ability between groups (concordance 0.80 for extinction, 0.85 for mining). These results suggest that larger worlds can sustain larger populations, thereby reducing the probability of reaching the absorbing zero-population boundary while simultaneously exploring a broader phenotype space through larger aggregate quantities of mutation.

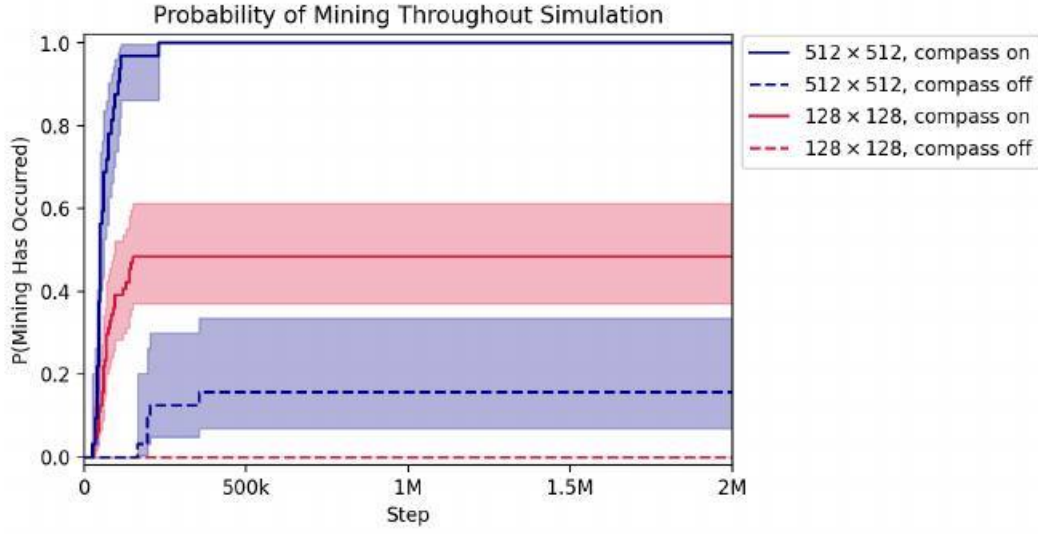


Figure 8: Probability that mining has evolved within the population, conditioned on world size, compass access, and timestep. We fit a separate non-parametric Kaplan-Meier estimator for each run group (e.g. 512×512 , compass on). Shaded regions denote 95% confidence intervals. Each 512×512 estimator is fit on 32 runs, and each 128×128 estimator is fit on 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events using the 128×128 , compass-off condition as the baseline. Hazard ratios (HR) below 1 indicate reduced hazard, corresponding to longer time to event; for example, extinction $HR = 0.04$ for 512×512 means that extinction occurs at 0.04 times the rate observed in 128×128 worlds. The analysis includes 192 total runs: 32 seeds with compass on and 32 seeds with compass off in the 512×512 world, and 64 seeds with compass on and 64 seeds with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	p-value	HR	95% CI	p-value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

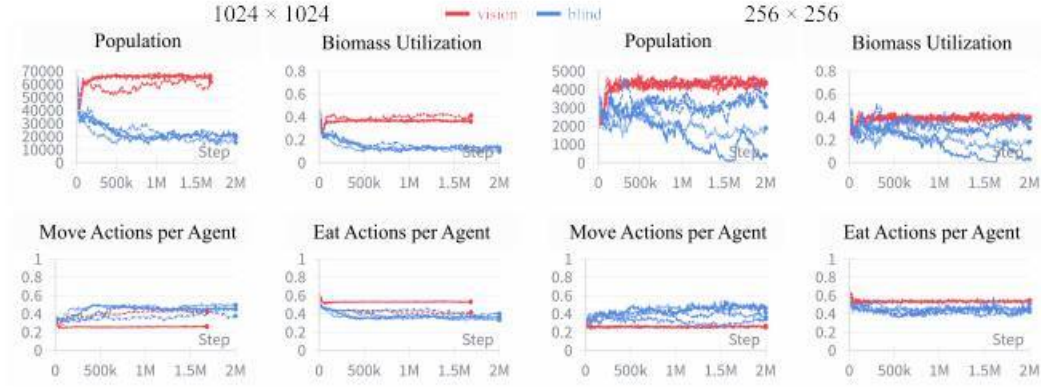


Figure 9: Foraging behavior in the **Ocean** world across scales. Enabling vision in agents (red) produces more stable populations and greater biomass utilization, driven by more frequent and effective eating together with reduced movement. At larger scales, these dynamics are more stable and exhibit lower across-run variance. Because of GPU-hour limitations, the vision seeds at the 1024×1024 scale were run for 1.7M rather than 2M steps.

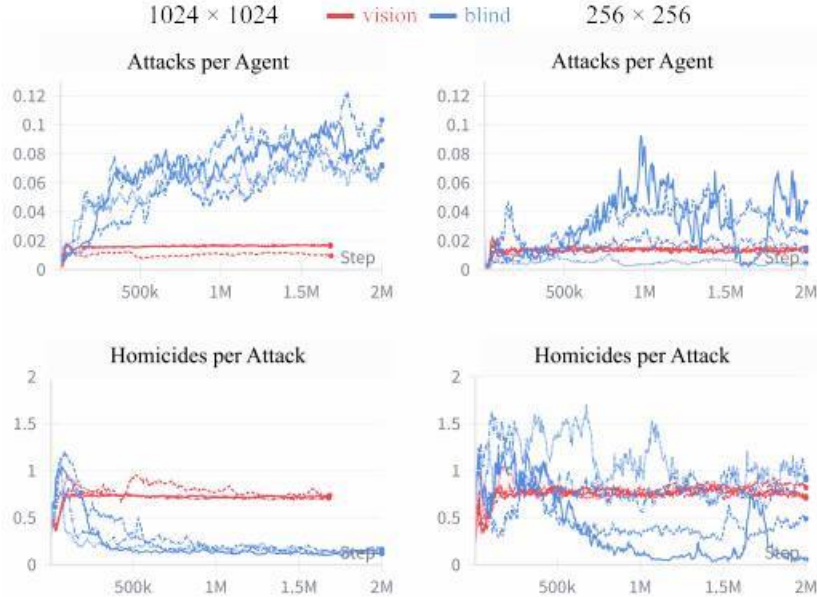


Figure 10: Predation behavior in the **Ocean** world across scales. With vision (red), agents attack less frequently but with much higher precision, suggesting the emergence of specialized predation behavior in addition to foraging capabilities (Figure 9). This dynamic is again more pronounced at larger scales, while blind agents (blue) exhibit greater behavioral variance in smaller worlds.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

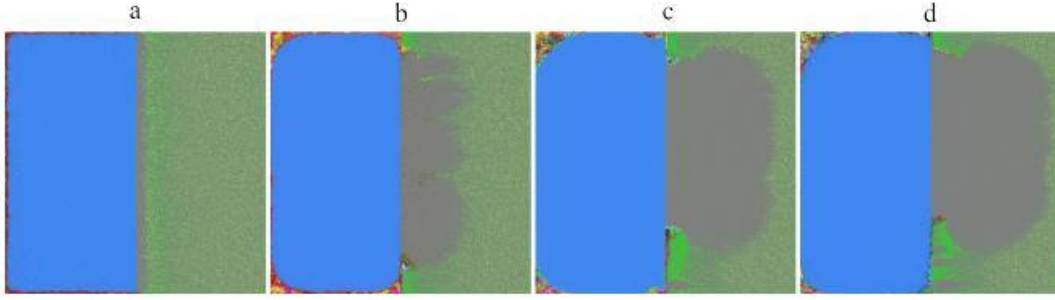


Figure 11: Larger-scale environments induce more varied and complex ecologies, producing distinctive visual patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially died at the shore (a), thereby transferring biomass onto dry land. They subsequently developed mining behavior and cleared biomass from the beach (b). When biomass re-accumulated at the ends of the beach (c), agent subpopulations re-developed mining behavior, traveling inland along these small biomass beaches to extract the resources (d).

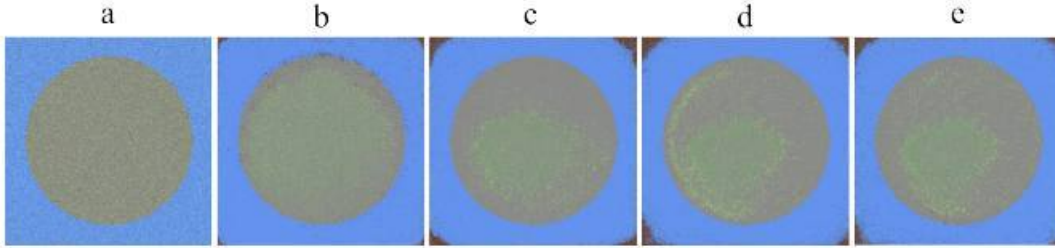


Figure 12: Mining behavior emerges across varied terrains. In this 512×512 **Island** world with **(RC)** agents, different shoreline regions are mined over the course of the run (b) (c). Once mining becomes difficult and agents die before returning to water, biomass is transferred back to parts of the island (d). Agents then adapt to re-mine this wall of accumulated biomass, clearing it from the shore (e).

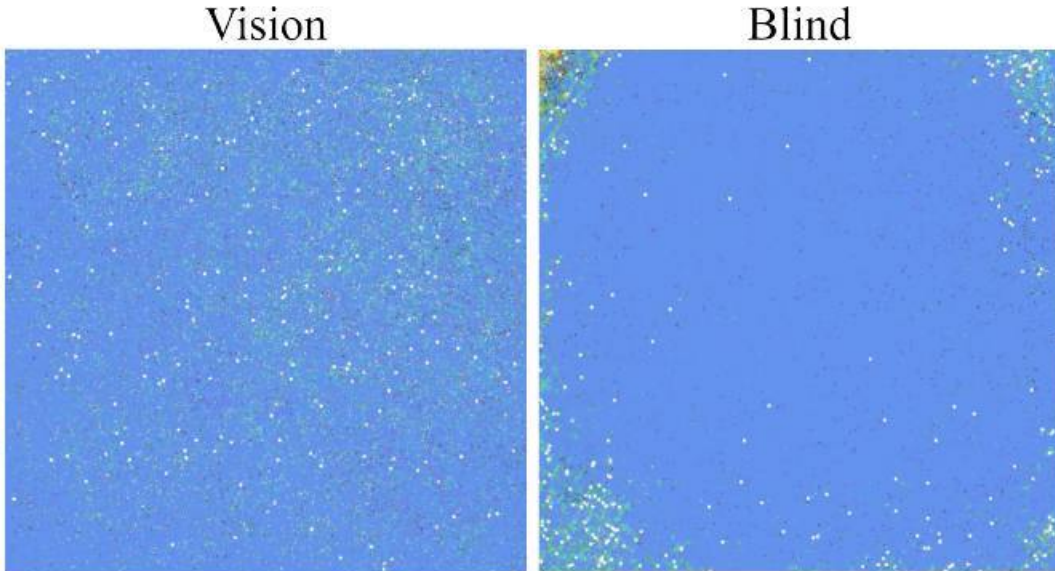


Figure 13: Representative runs in a 512×512 **Ocean** world with attacking enabled, showing visual **(RCV+A)** agents on the left and blind **(RC+A)** agents on the right. White squares indicate locations at which agents are attacking. With vision, agents adapt to locate free resources throughout the ocean world and prey on other agents more effectively (see Figures 5 and 6). Blind agents explore the ocean for resources to a much lesser extent, instead adapting policies that cluster in the world corners.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We provide additional details concerning simulator design, runtime, and experimental reproducibility.

Simulator design. All experiments are conducted with a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every component of an environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation execute within a single JAX compilation pipeline. It exposes a minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits while handling births, deaths, and environmental transitions. This separation permits the same environment implementation to be reused with evolutionary algorithms, reinforcement-learning methods, or other population-update rules. All operations are implemented as pure JAX functions composed through vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This design provides the throughput required for our scaling experiments, with single-H100 runtime measurements reported in Table 8.

A central feature of the simulator is its built-in tooling, including a measurement API that computes and records aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters at each step, all of which are used directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools were essential for discovering and validating qualitative phenomena, including long-range resource mining and predation patterns, that are not readily apparent from scalar metrics alone.

From a high-performance-computing perspective, the simulator addresses a gap left by existing platforms. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), or Griddly (Bamford et al., 2020) provide substantial configurability, but they require considerable environment engineering and do not directly target open-ended, evolving populations at the scales considered here. Conversely, GPU-native multi-agent RL environments typically emphasize fixed agent counts and task-specific reward structures. Our simulator combines GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination provides a state-of-the-art platform for scaling ecological experiments such as those reported in this paper, as well as a useful foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime measurements across grid sizes and maximum agent populations. Each run was executed on a single H100 GPU for at most 2M steps. Although the table reports run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), runtime depends on agent-network size, available sensors, and logging intensity.

At the 1024×1024 grid size, our largest runs require ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on a single H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower because they include both the agent attack action and vision sensing.

Reproducibility. Because the environments are large and randomized, precise experimental outcomes can vary substantially across runs. Accordingly, we report four random seeds for each experiment. Due to nondeterminism in some low-level GPU operations, however, repeating an experiment with the same random seed does not reliably reproduce an identical trajectory. In particular, modern GPUs may execute parallel floating-point operations, such as summations, in different orders across runs and may fuse operations in ways that alter rounding behavior. Minute differences induced by finite-precision arithmetic can therefore accumulate over many steps into noticeable simulation drift. The use of 16-bit weights can coarsen these rounding effects, although it is not the primary source. Despite run-level variation, clear high-level patterns emerge when multiple seeds are considered in aggregate.

Table 8: Runtime measurements in the **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each **(RC)** agent has a 22,465-parameter neural network, and each **(RCV + A)** agent has a 22,529-parameter neural network. The table reports simulation steps per second (hz) and extrapolated wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds in which populations go extinct before 2M steps.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

F ABLATIONS

We present a series of ablations investigating the effects of specific hyperparameters and robustness of results under different environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate the effect of initial population size in a 256×256 **Beach** world on the consistency of mining behavior and population stability. Results reported are for **(RC)** agent populations. Agents without compass **(R)** had 0% mining rate and 100% extinction rate across all initial population sizes except 8192, the highest tested, with 25% mining and 75% extinction frequency. Note that mining and extinction frequencies need not add to 100%: it can be the case that a population develops mining but then goes extinct, for instance, or that a population lasts for the full 2M steps of our experiments without developing the mining behavior.

In Figure 14 we find that the initial population size has no significant effect on mining consistency or extinction rates: populations starting with only 128 agents evolve mining behaviors as frequently as populations starting with 8192 agents, and extinction rates are largely stable across initial population size.

Figure 15 provides some insight into why so little effect is visible. Regardless of the size of the initial population, each run rapidly converges to a similar population size in the first 50,000 steps of simulation. This suggests that the environment implicitly specifies a carrying capacity based on the fixed resources available in the system: populations beginning below the carrying capacity grow to reach it, and populations that are too large are constrained to the capacity. As a result, the effective population size is similar after an early correction of the initial population size, leaving mining and extinction frequencies unaffected. Importantly, however, larger initial population sizes contribute somewhat to the total resources in the system because agents carry biomass that is deposited upon death; as a result, larger initial population sizes are able to reach higher maximum populations on average over the course of simulation (Table 9), but the effect is not significant enough to impact mining or extinction frequency for this setting. It is worth noting that although the population size converges early on, it does not stay consistent throughout the run. Later in the simulation, the populations between different runs can vary substantially, as the ecological impacts of agents evolving different abilities at different times changes the later resource distributions.

F.2 VARYING RESOURCE RULES

We evaluate the effect of two resource hyperparameters, biomass photosynthesis rate and initial biomass site density, on mining and extinction frequency. The setting is equivalent to that in Appendix F.1. The initial population size is fixed at 2048 for this study, and agents once again have compass **(RC)**.

The biomass photosynthesis rate controls the rate at which landscape biomass creates energy via photosynthesis, providing resources for agents over time. The initial biomass site density parameter determines the fraction of grid cells that have biomass at the start of simulation. All experiments

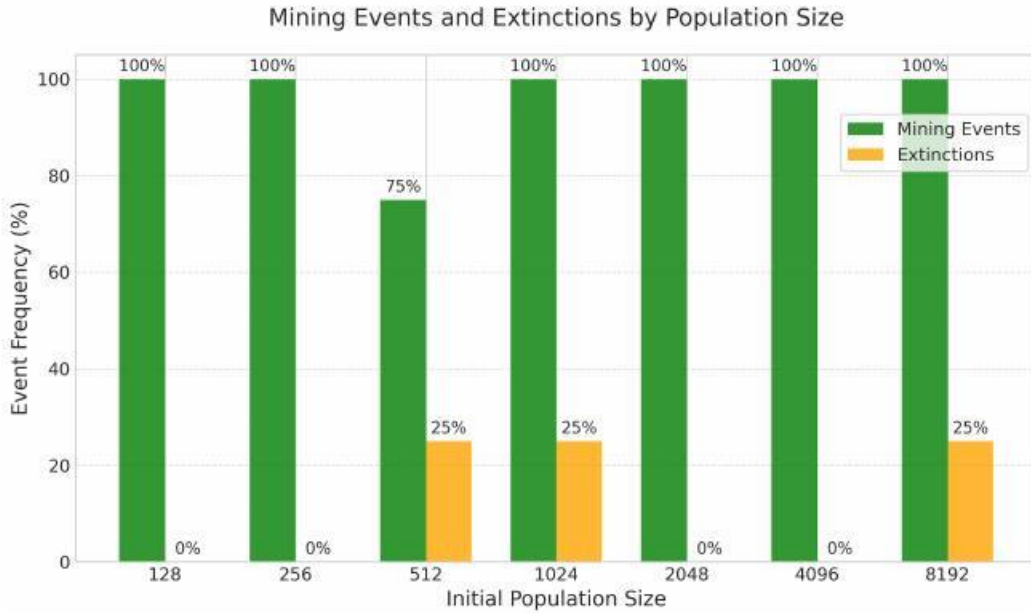


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world for different initial sizes of (**RC**) agent populations. Frequencies are computed across four random seeds for each initial population size; for example, 25% Extinction Event Frequency indicates that in 1/4 seeds, the population reached 0 before 2M simulation steps.

Table 9: Maximum population size reached after step 0 for different starting population sizes, reported as averages and standard deviations across four random seeds. Larger initial populations reach slightly higher maximum populations and exhibit greater seed-level variance than smaller initial populations. Nevertheless, maximum population size is largely robust to initial population size: an initial size of 128, only 1.6% of 8192, reaches 78.9% of the maximum population reached by the latter condition (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

While varying initial population size did not affect mining and extinction consistency (Figure 14), changing resource rules has a significant effect as demonstrated in Figures 16 and 17.

Intuitively, larger biomass photosynthesis rates or initial biomass site densities directly boost the effective carrying capacity of the environment by increasing the resources available in the system. As a result, larger populations are supported throughout the run, increasing the likelihood of mutations favoring mining behavior while lowering the chance of extinction. Our data support this intuition: on average (omitting seeds that went extinct), runs with initial biomass site density 1/16 reached a max population of 1608 after step 0, while reaching a mean max population of 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

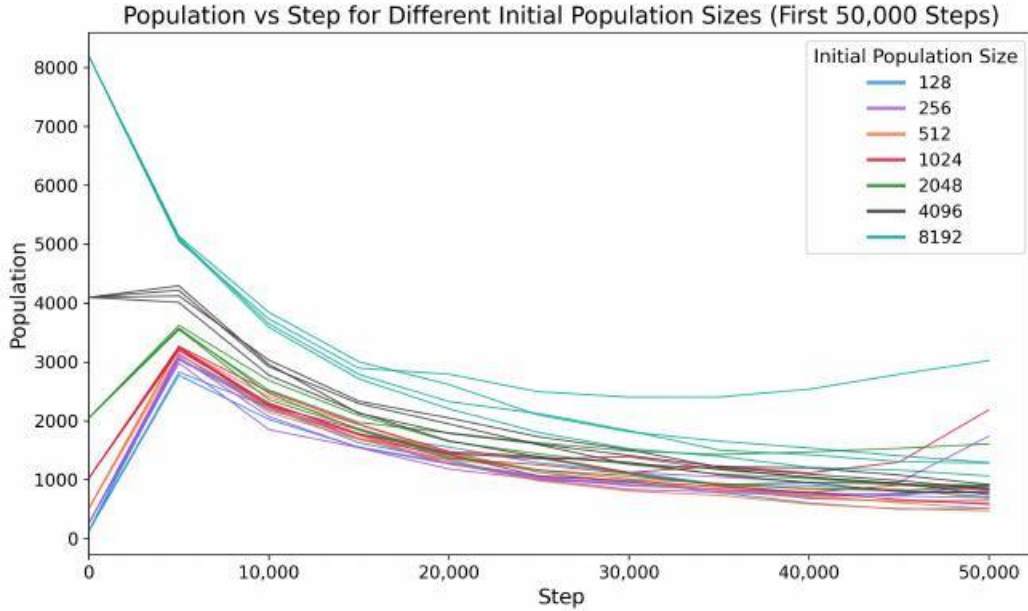


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** world with **(RC)** agents, using four seeds per initial population size. The environment imposes an implicit carrying capacity determined by available system resources, and all initial population sizes approach this value early in the run. Larger initial populations contribute additional total resources through biomass deposited after agent death, thereby modestly increasing carrying capacity.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One concern is that the increased stability of mining behavior in large environments (Table 1) might be attributable purely to self-averaging rather than to qualitatively distinct dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, such that a single large system behaves like the average of smaller systems of the same total size. Under this view, a 512×512 **Beach** world might simply be acting like many loosely coupled 128×128 worlds run in parallel. If that were the case, then our scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

To test this hypothesis, we run an experiment that holds the total “sample size” approximately fixed while varying how that sample is partitioned across environments. In the **Beach** terrain with compass-equipped agents **(RC)**, we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1. As a control, we repeat this protocol for agents without a compass **(R)**. Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

If large environments merely acted as self-averages of many smaller ones, mining and extinction rates should be approximately invariant to the distribution of grid cells and agents across runs: 4

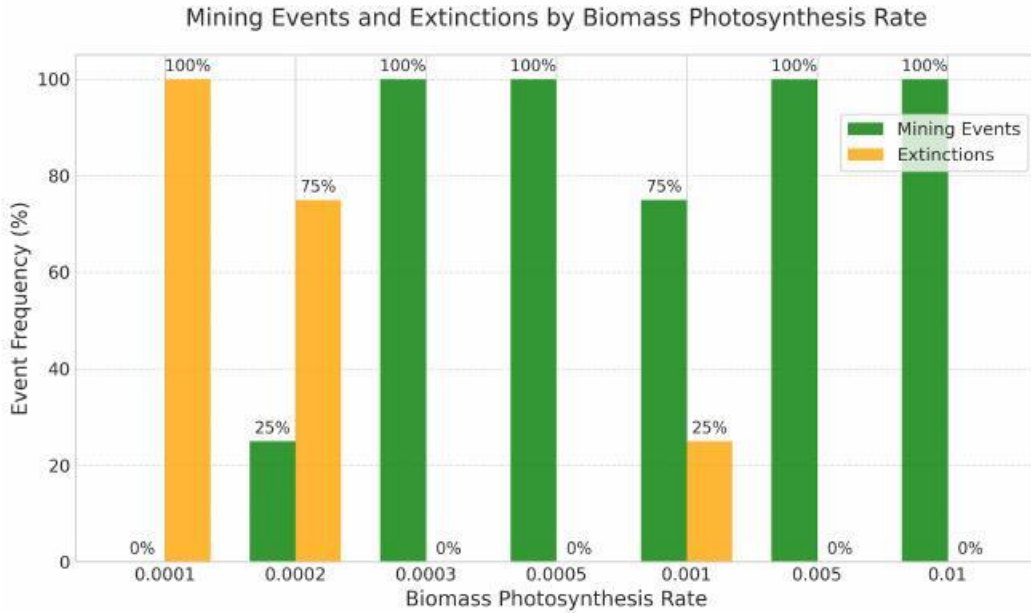


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world under different settings of the biomass photosynthesis rate resource parameter. Frequencies are computed across four random seeds for each setting. Mining consistently emerges when biomass photosynthesis rates are 0.0003 or greater.

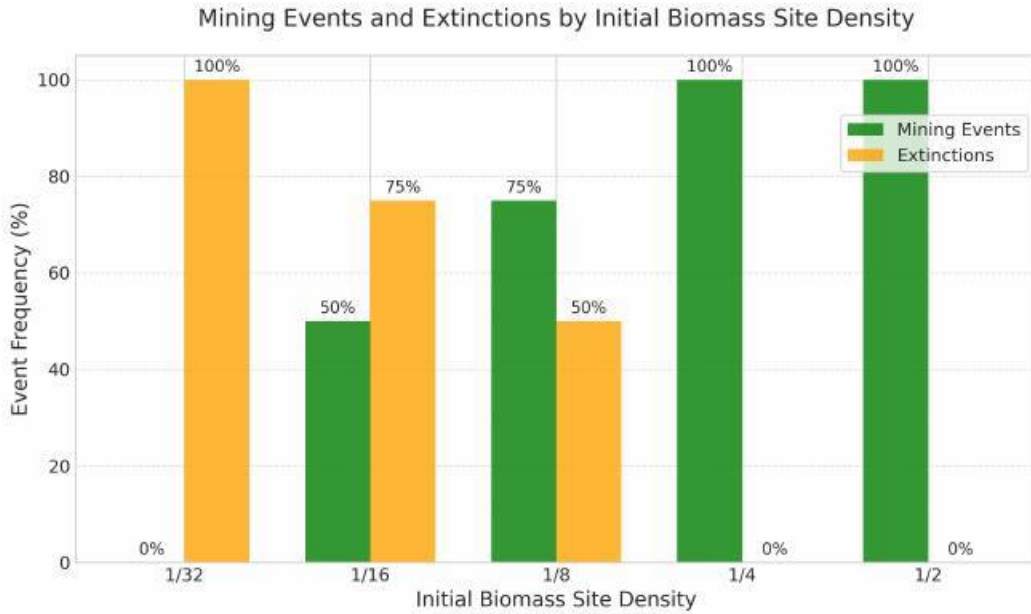


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world under different settings of the initial biomass site density resource parameter. Frequencies are computed across four random seeds for each setting. Greater initial biomass site densities yield more consistent mining discovery and lower extinction probability.

large worlds or 64 small ones would exhibit similar probabilities of discovering mining. Instead, we observe a strong dependence on world size. For (RC) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable

and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, **(R)** agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

Table 10: Mining and extinction events when trading world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , and 128×128 , with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, the same total number of grid cells and initial agents is allocated across all runs by using 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged () and the fraction in which the population went extinct before 2M steps () for agents with a compass (**(RC)**) and without a compass (**(R)**).

	512×512		256×256		128×128	
	8192		2048		512	
						
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are inconsistent with a self-averaging explanation. On the contrary, larger environments appear to enable and stabilize mining behavior in ways that cannot be explained solely by averaging over more small simulations.

Even when matching the total number of cells, agents, and simulation steps across conditions, concentrating resources into a few large worlds produces a regime in which mining is consistent and extinctions are rare, while dispersing the same resources across many small worlds gives populations that frequently fail to discover mining and are more likely to collapse. This suggests that increasing environmental scale is not simply smoothing out stochastic noise, but is altering the underlying ecological dynamics, for example by supporting larger, more spatially distributed populations and more persistent diversity in evolving policies.

F.4 MUTATION RATE SWEEP

We test the sensitivity of the emergence of mining to the mutation rate hyperparameter, which controls the scale of Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate directly impacts how different a child agent is from its parent. We once again run experiments in a 256×256 **Beach** world with an initial population of 2048 **(RC)** agents for a maximum of 2M steps.

In Figure 18 we show that between mutation rates of 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds run, indicating that our results in Section 4.1 with mutation rate 3×10^{-2} are robust within this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction. Conversely, large mutation rates (e.g., 3×10^{-1}) undermine population stability, consistently leading to extinction with no mining adaptation.

F.5 NETWORK SIZE

We study the effect of the size and architecture of the agent policy network on mining and extinction frequency, as well as the impact of environmental scale on this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 **(RC)** agents, respectively (scaling by four with the number of grid cells). We ran baselines with **(R)** agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, yielding a network of 22465 parameters per agent (Table 11). In Figure 19 we show that in the 256×256 **Beach**, halving channels to 32 (resulting in 9185 total parameters) or doubling to 128 (giving 61313 parameters) with number of layers fixed has minimal to no effect on mining and extinction frequency. Meanwhile, changing network depth has a significant effect: all populations of agents with just 1

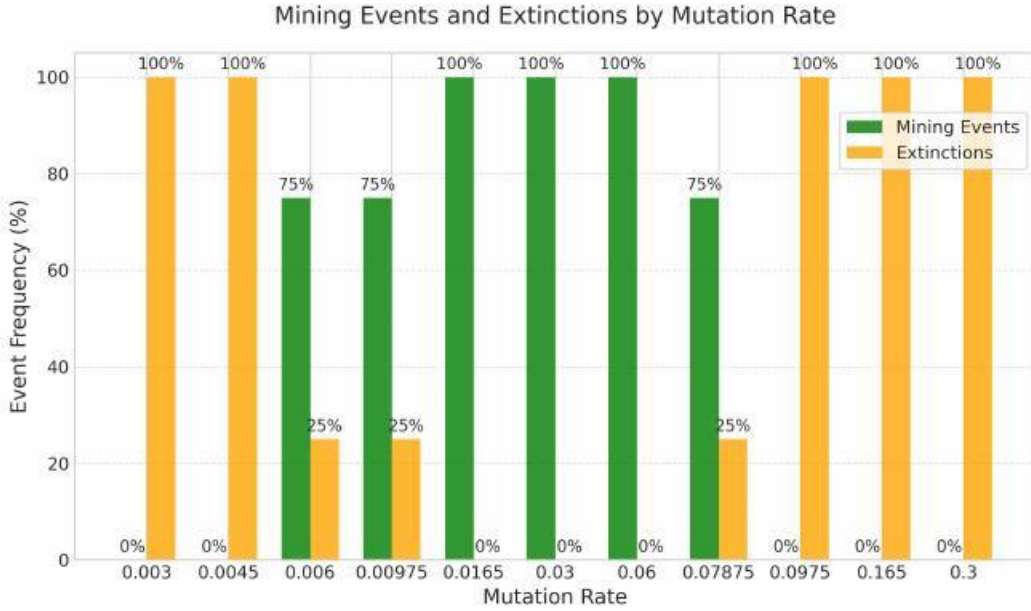


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world under different mutation rates. Frequencies are computed across four random seeds for each setting.

layer and 32 channels went extinct, and 50% of populations of agents with 3 layers and 128 channels went extinct.

Crucially, scaling the environment to the 512×512 grid size produced significantly different dynamics for networks of varying depth (Figure 20). In particular, in the larger world, agent populations with 3 layer, 128 channel networks consistently mined and survived through 2M steps. Surprisingly, agent populations with the small 1 layer, 32 channel networks mined consistently as well. We hypothesize that it is more challenging for both smaller and larger networks to discover new behaviors for different reasons. When the network is smaller, the reduced capacity makes it hard to find good solutions. When the network is larger, the increased size of the search space means it takes longer to find promising new behavior. Regardless of the reasons, the fact that these networks fail so often in small environments but succeed in larger ones further illustrates the stability that comes from increasing scale. The larger environment can support larger populations, which allows for more exploration in policy space. Simultaneously, it takes longer for locally successful mutants to propagate across a large environment, which promotes diversity and again ensures greater exploration. For more discussion of these effects see (Mitchell et al. 2006).

Table 11: Total parameter counts for (RC) and (R) agent policies under different network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

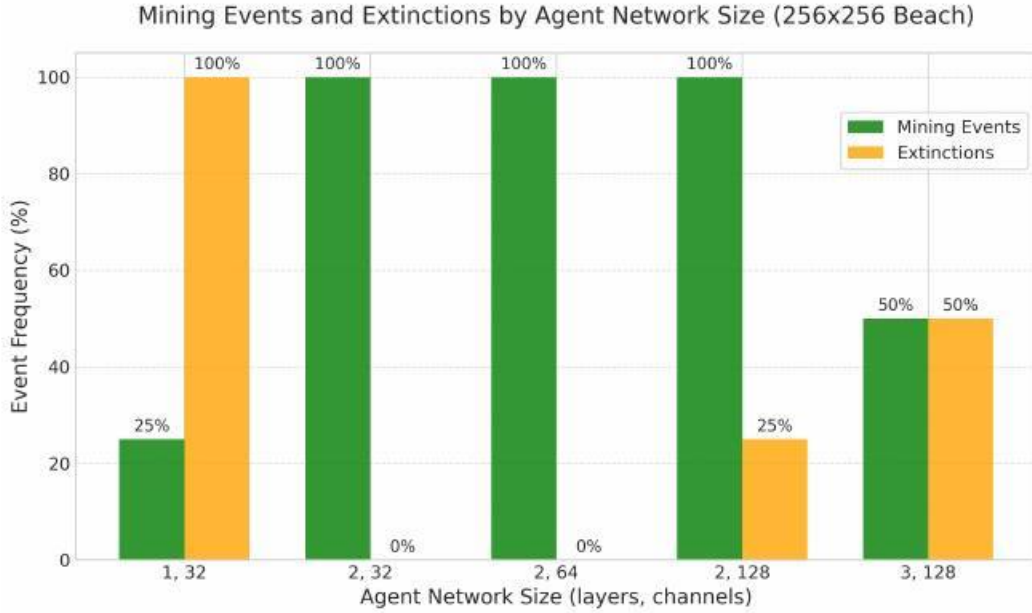


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world under different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

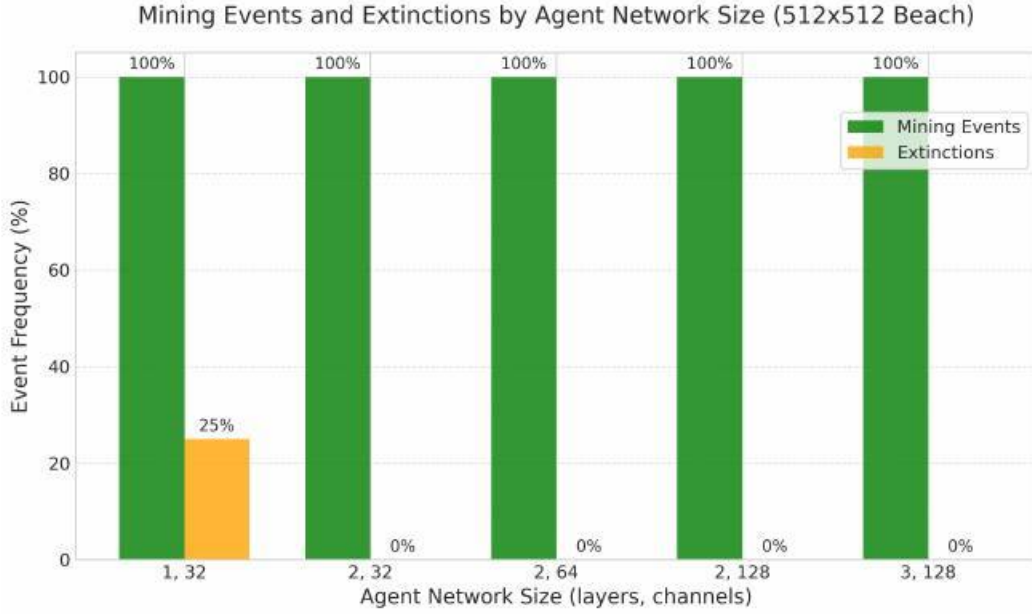


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world under different agent policy network sizes. Frequencies are computed across four random seeds for each setting.

F.6 VARYING VISION RANGE

We assess the impact of different vision ranges and viewing configurations on predation and foraging behaviors in the 512×512 **Ocean** world with (**RCV+A**) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 . For each range, we evaluate two viewing configurations shown in Figure 21 a) Around: agents can see grid cells all around them b) Front: agents cannot see cells behind them.

The total number of cells in the agent's vision range is equal for the two configurations. The main experiments with vision in Section 4.2 were run with the 7×7 Around setting.

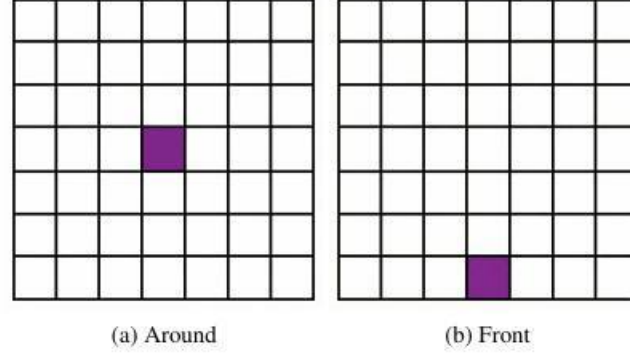


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell represents an agent facing the top of the page.

Table 12: Foraging and predation data in the 512×512 Ocean world for visual agents (**RCV+A**) with varying vision ranges and viewing configurations. Tables 5 and 6 define the metrics represented by each icon. For each seed, each metric is averaged across all steps beginning at step 500k to allow stabilization. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

These results indicate that the smaller 3×3 vision range produces behavior largely similar to the no-vision condition (Blind), whereas the 9×9 results appear similar to those from 7×7 . This indicates that there may be diminishing returns of larger vision ranges in the current environmental configuration. This is intuitive given the rest of the environmental configuration; since the agents have no memory, it is unlikely that visual features at the periphery of a 9×9 window could meaningfully alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

The data also suggest that, at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable with respect to population stability and the effectiveness of attack and eat actions, resulting in higher average population , biomass utilization , and homicides per attack  than the Front configuration. We hypothesize that this is due to diminishing returns of being able to see more distant pixels; in the Around configuration, more pixels in the agent's vision range are nearby, likely containing more relevant information for near-term actions than distant cells visible with the Front configuration. At the 3×3 vision range, however, the trend is reversed, with the Front configuration being more favorable than Around; this indicates that in this environment, the marginal benefit of seeing one more pixel ahead through the 3×3 Front configuration outweighs that of seeing one pixel behind in the 3×3 Around configuration, which gives most similar behavior to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We examine how reducing agent attack strength affects emergent behavior and population dynamics in the 256×256 **Ocean** world. We evaluate the following attack strengths: 1) Full Attack: deals 10 HP of damage, instantly killing any agent in the attack radius 2) Half Attack: deals 5 HP of damage 3) Quarter Attack: deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

We find that reducing attack strength from Full Attack to Half Attack substantially decreases both the effectiveness and frequency of attacks: attacks per agent and homicides per attack drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently, resulting in greater biomass utilization. Further reducing strength to the Quarter Attack setting yields diminishing returns in terms of further reduction in violence and facilitation of additional population growth. Intuitively, when attacks do less damage, a homicide requires more attacks per agent, reducing the lethality of each attack. However when the lethality is reduced, attacking behavior becomes less useful and therefore less likely to be selected for in the agent population. Agents with lower attack strengths therefore do not become as skillful at attacking and spend more time foraging for food.

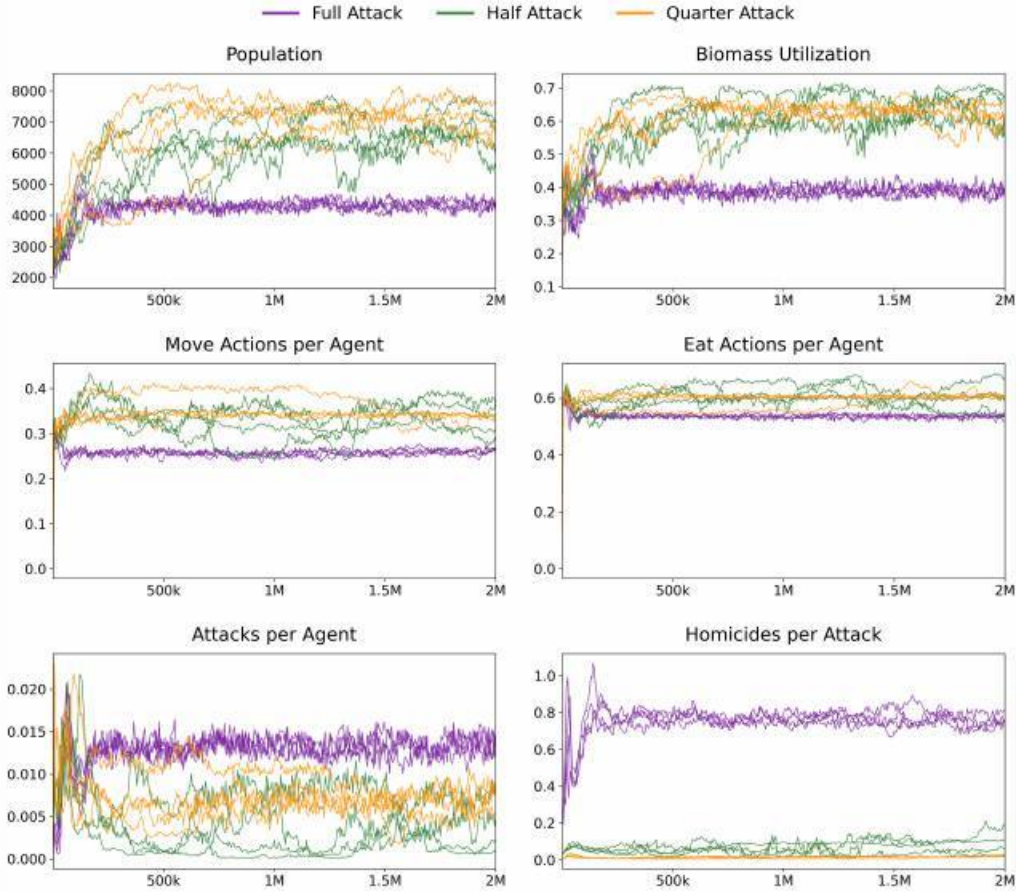


Figure 22: Population and action-frequency data over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted for each attack-strength setting.

F.8 VARYING AGENT HP

We investigate the effect of reducing child agents’ starting HP and lowering the healing rate, thereby imposing pressure for agents to ‘grow’ over their lifetimes in order to survive. The healing rate parameter determines how much HP an agent can regenerate per healing step, which converts available energy to HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, we observe no substantial difference in homicides per attack, eat-action frequency, or biomass utilization (Figure 23). In the Full HP setting, populations grow slightly larger, while in the Low HP setting, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must focus on foraging early on in their lifetime in order to grow to full size, which would explain this difference.

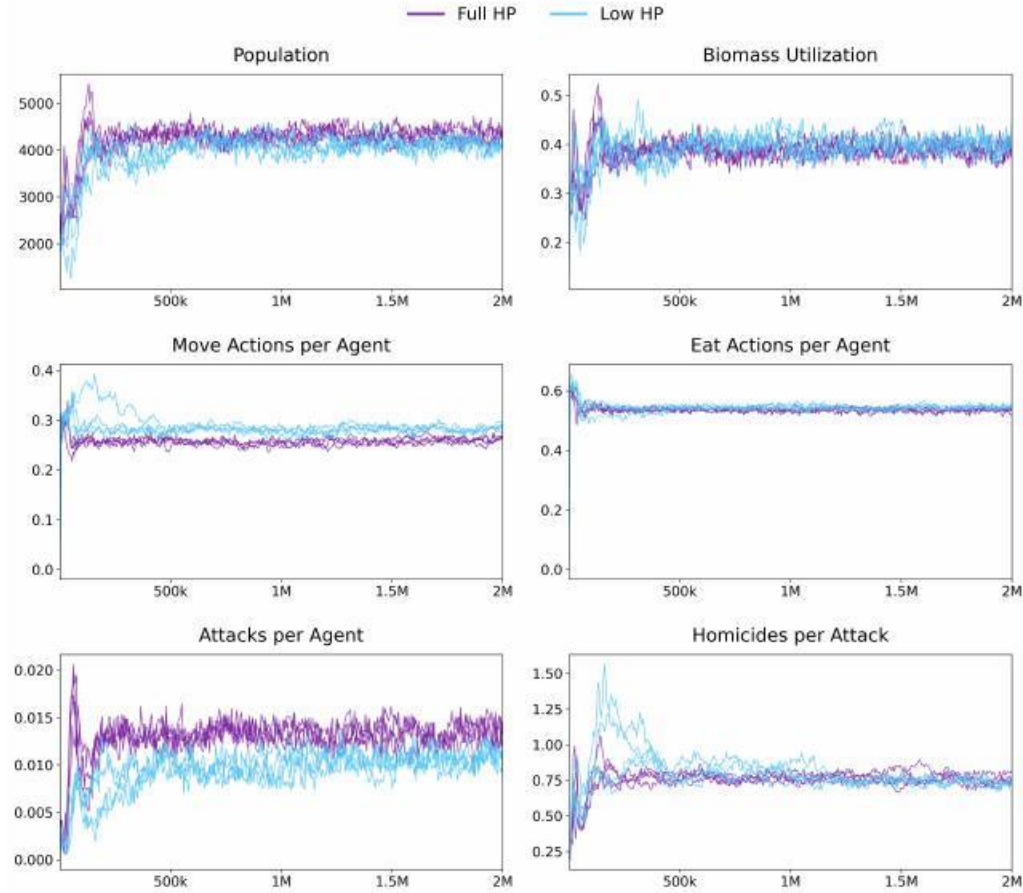


Figure 23: Population and action-frequency data over 2M steps in the 256×256 **Ocean** world for (RCV+A) agents across Full HP and Low HP settings. Four seeds are plotted for each HP setting.